

Advanced language features 2:

SPLIT, EXTEND, WHEN, COPY & GROUP

(... and the dangerous JUMP)





Syntax in one, complex view...

{**SPLIT**} COMPONENT name = comp(parameters) {**WHEN** condition}
AT (...) [RELATIVE [reference|PREVIOUS] | ABSOLUTE]
{ROTATED {RELATIVE [reference|PREVIOUS] | ABSOLUTE} }
{**GROUP** group_name}
{**EXTEND** C_code}
{**JUMP** [reference|PREVIOUS|MYSELF|NEXT] [**ITERATE** number_of_times | **WHEN** condition] }

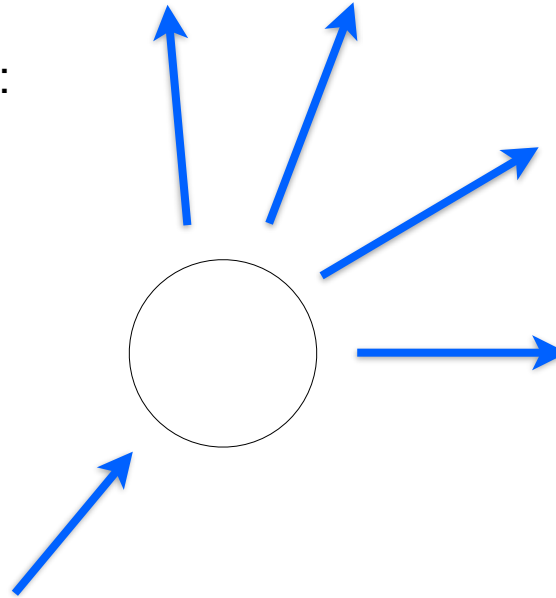
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SPLIT

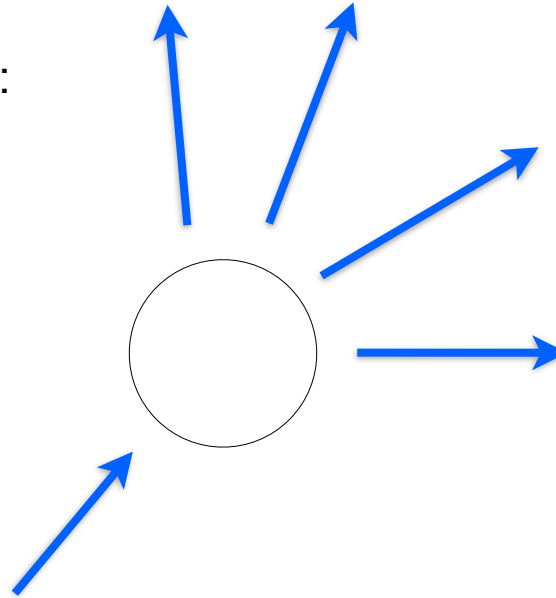
- Increase statistics beyond this point in the instrumentfile
- `SPLIT n MyArm = Arm()`
- AT somewhere
- will “formulate an if-statement”:
 - for `j=1:n`
 - `comp1`
 - `comp2`
 - `comp3`
 - ...
 - end (of instrument)
- ONLY meaningful in case of Monte Carlo choices after SPLIT point...





SPLIT

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Problem: McStas Single_crystal.comp “slow” for large unit cell diffraction studies

- Example: Rubredoxin

1 timebin, 1000 x,y-bins

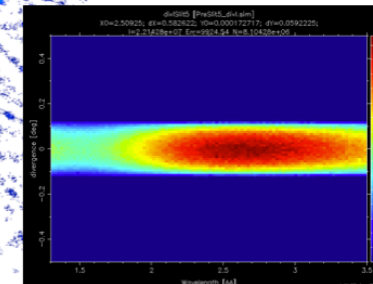
Rubredoxin

Images created from simulated datafile produced
August 20th 2012 using 25 nodes on the DMSC
cluster.

Neutron count: 1e12

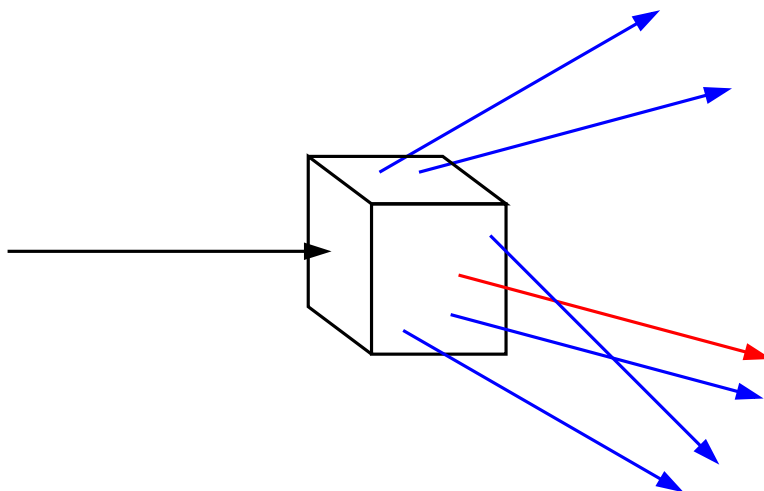
Simulation time: ~10 + ~20 hr = ~30 hrs total

Neutroncount: 1e12
No gravitation
Xtal size: 0.5 mm
Xtal mosaicity: 12'
Detector: 50 x 50 cm flat
Detector-to-sample
distance: 20 cm
Guide length: 131 m
Guide dimensions: 9.5 cm
 $\lambda_{\min} = 1.3 \text{ \AA}$
 $\lambda_{\max} = 3.5 \text{ \AA}$
Timespan: 51.39 to 143.4 ms
Divergence = 0.2 degs



Algorithm improvement: Use incoming neutrons more efficiently - scatter each one on all possible reflections

- **Red**: Original algorithm, one incoming neutron used only once
- **Blue**: Improved algorithm, each incoming neutron scattered (via SPLIT keyword) all possible times
- Component makes **estimate on average number of “active” diffraction spots** - in the case Rubredoxin this is around **50!**





GROUP - components working in parallel



*AT (0,0,-LMM) RELATIVE Cradle ROTATED (0,A1/2,0) RELATIVE Cradle
GROUP IN6Monoks*

*AT (0,0,0) RELATIVE Cradle ROTATED (0,A2/2,0) RELATIVE Cradle
GROUP IN6Monoks*

*- One comp after the other is “tried” in sequential order until the neutron was
SCATTERED.*



EXTEND

- Enrich component behaviour using EXTEND:

```
COMPONENT Mono1 = Monochromator_curved(...)
AT (0,0, -LMM) RELATIVE Cradle ROTATED (0,A1/2,0) RELATIVE Cradle
GROUP IN6Monoks
```

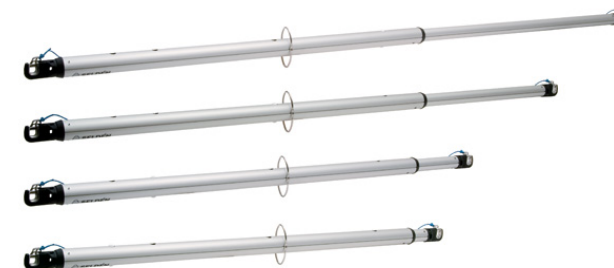
EXTEND

```
%{
  if (SCATTERED) { myvar = 1; }
%}
```

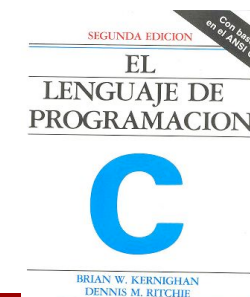
...

```
COMPONENT Mono2 = Monochromator_curved(...)
AT (0,0, 0) RELATIVE Cradle ROTATED (0,A2/2,0) RELATIVE Cradle
GROUP IN6Monoks
```

```
%{
  if (SCATTERED) { myvar = 2 ;}
%}
```



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WHEN

- Syntax:

COMPONENT Mine = Yours(blah, blah)

WHEN (c-expression) AT (....)

- Is very powerful when combined with EXTEND and user variables, or as a method to let input parameters select if certain components are active.

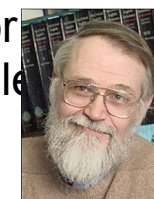
- Example: Use EXTEND to flag if neutron was scattered on one monochromator another. Then later use WHEN to only show contribution from blade N at sample

COMPONENT Mon = PSD_monitor(...)

WHEN (myvar==1) AT (0,0,0) RELATIVE Sample



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JUMP

- A goto. Be careful. Can be used in two situations:
- JUMP to myself
- JUMP to an Arm
- No coordinate transformations are applied... (Meaning that if the Arms you JUMP between do not coincide you will “move” / “reorient” the neutrons...)
- Syntaxes:
- COMPONENT a=b(...)
- WHEN (expr) AT (...) JUMP somewhere
- COMPONENT a=b(...)
- WHEN (expr) AT (...) JUMP myself



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BEWARE - This IS a



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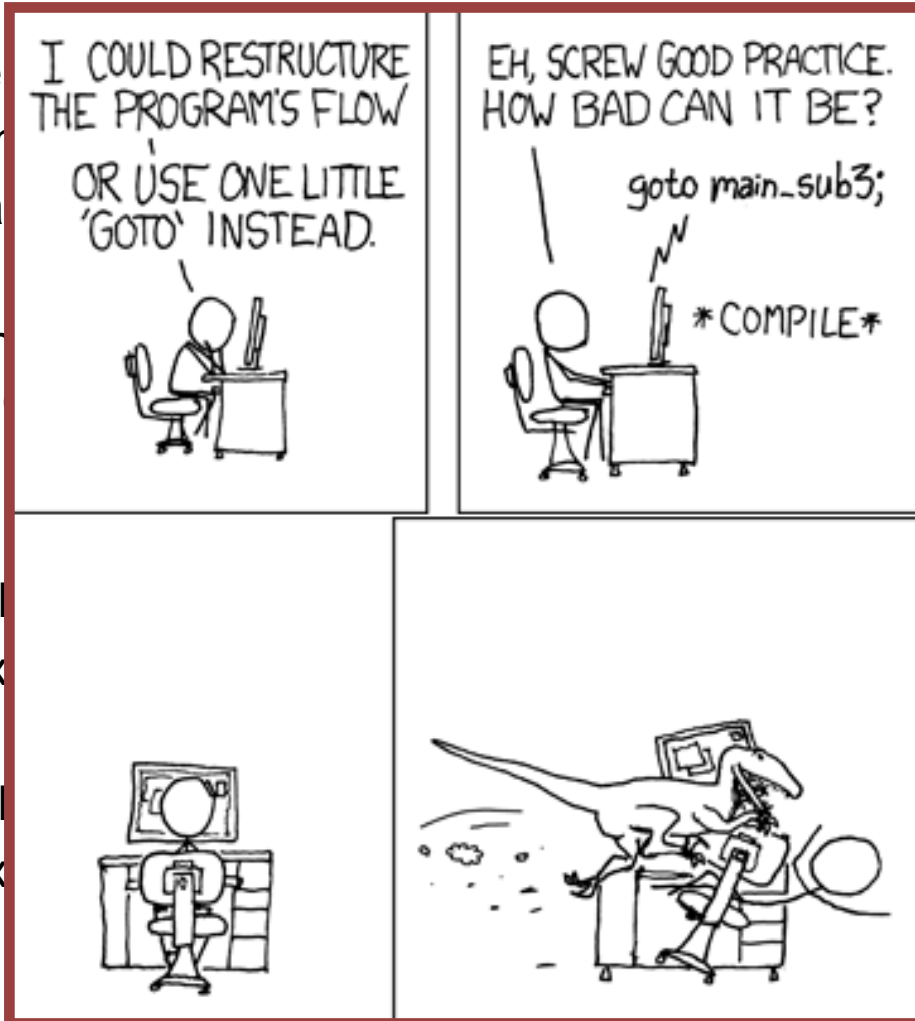
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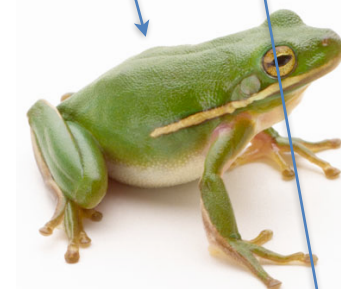


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COPY- inside instruments

- In instruments: (see ILL_H25.instr)
- COMPONENT H25_1 = Guide_gravity(
 - $w1=0.03$, $h1=0.2$, $w2=0.03$, $h2=0.2$, $l=L_H25_1$,
 - $R0=gR0$, $Qc=gQc$, $\alpha=g\alpha$, $m=m$, $W=gW$)
 - AT (0,0, $A_Thickness+gGap$) RELATIVE PREVIOUS
 - ROTATED (0, $Rh_H25_1,0$) RELATIVE PREVIOUS
- COMPONENT COPY(H25_1) = COPY(H25_1)
 - AT (0,0, L_H25_1+gGap) RELATIVE PREVIOUS
 - ROTATED (0, $Rh_H25_1,0$) RELATIVE PREVIOUS
- COMPONENT COPY(H25_1) = COPY(H25_1)($W=2*gW$)
 - AT (0,0, L_H25_1+gGap) RELATIVE PREVIOUS
 - ROTATED (0, $Rh_H25_1,0$) RELATIVE PREVIOUS



Example: Decompose multiple scattering from Single_crystal

```
DECLARE %{\n  double multiple_scatt;\n}%\n...\nCOMPONENT Crystal = Single_crystal(... order=0 ...)\nAT (0,0,0) RELATIVE somewhere\nEXTEND %{\n  multiple_scatt=SCATTERED;\n}%\n...\nCOMPONENT PSD_single=PSD_monitor(...)\nWHEN (multiple_scatt==1) AT (0,0,0) RELATIVE somewhere_else\n\nCOMPONENT PSD_multiple=PSD_monitor(...)\nWHEN (multiple_scatt > 1) AT (0,0,0) RELATIVE somewhere_else
```

